

Digital Twin Final Project

Isaac Sim · Isaac Lab — Extending your midterm with robot learning

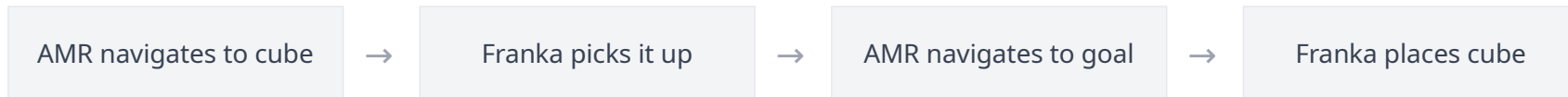
Imitation Learning

**Reinforcement
Learning**

Project Overview

The midterm pipeline — now with learned policies replacing scripted behaviors

MIDTERM PIPELINE



REPLACE SCRIPTED BEHAVIORS WITH LEARNED POLICIES

IMITATION LEARNING

Franka Pick & Place

Collect expert demonstrations, then train a policy to pick and place the cube with position and orientation precision across three difficulty levels.

REINFORCEMENT LEARNING

Nova Carter Navigation

Train an AMR agent to navigate from start to goal — heading-aware and optionally with static obstacle avoidance.

1 — Training Environments

40 points · You build these yourself — TA provides eval config only

Environment	Purpose	Points
RL training environment	AMR navigation training	20
IL data collection environment	Expert rollout + dataset generation	20

 Submitting only the environment class without training evidence = partial credit at best.
Your README must include exact commands to reproduce both environments.

README must include:

- ☐ Python + Isaac Lab version, any extra dependencies
- ☐ Exact command to run IL data collection and training
- ☐ Exact command to run RL training

2 — Trained Policies

50 points · Pass threshold: $\geq 80\%$ success rate over evaluation episodes

IMITATION LEARNING — Franka Pick & Place

Level	Task	Pts
L1	Pick (arbitrary yaw), place at target Position error < 5 cm	10
L2	Extend L1 + specified orientation Yaw error < 15°	15
L3 ★ Bonu s	Pick from randomized-height platform, place at different height with orientation	+5

REINFORCEMENT LEARNING — Nova Carter Navigation

Level	Task	Pts
L1	Navigate start to goal Position error < 10 cm, straight path	10
L2	Extend L1 + specified heading Yaw error < 15°	15
L3 ★ Bonu s	Extend L2 + static obstacles Avoid all collisions Obstacle assets provided — evaluation draws from that set only	+10

3 — Report & Demo Video

10 points · PDF max 8 pages + recorded demo in Isaac Sim viewer

PDF REPORT

- Scene & environment design — assets, sensors, differences from midterm
- IL pipeline — expert policy, demo count, hyperparameters, loss curves
- RL pipeline — obs / action / reward design with rationale, training curves
- Domain randomization — what, ranges, screenshots of variations
- Results — success rate table per level with screenshots / video frames
- Discussion — what worked, what did not, future directions

DEMO VIDEO

- Record in Isaac Sim viewer
- Show all IL levels attempted
- Show all RL levels attempted
- Label each clip with the level name

Both IL and RL tasks must be demonstrated in the video. Label each segment with its level name for clarity.

Evaluation Protocol

What the TA provides vs. what you submit alongside your trained checkpoints

TA PROVIDES

EvalSceneCfg — scene configuration
sensor_setup_fn(env) — hook for extra
sensors

YOU PROVIDE

Trained checkpoints
observations — terms your model expects
actions — action space definition

GROUP PROJECT

This is a group project.

Each group consists of 3-4 members. A Google Sheet link will be provided for you to register your group and fill in member details.



GROUP SHEET



Link provided
via course page

3-4 members per group

Name · Student ID · Email

Submission

Fill in the FTP details and deadline below before distributing to students

DEADLINE

June 8, 23:59

FTP Upload Path

DigitalTwin2026_HW/Final

FILENAME FORMAT

<group_id>_final_project.zip

 Files on the FTP server cannot be deleted or overwritten. To resubmit, append a version number: <group_id>_final_project_v2.zip — only the latest version will be graded.

POINT SUMMARY

Training environments

40 pts

Trained policies

50 pts

Report + video

10 pts

Bonus available

+15 pts

What Goes in Your ZIP

Filename: <group_id>_final_project.zip — everything the TA needs to run and grade your work

ENVIRONMENTS

rl_env/

RL training environment class

il_env/

IL data collection
environment class

README.md

Setup, dependencies, exact
run commands

CHECKPOINTS

il_policy.*

Trained IL model (all levels
attempted)

rl_policy.*

Trained RL model (all levels
attempted)

policy_cfg_L*.py

Per-level config (obs +
actions), one file per level
attempted

REPORT & VIDEO

report.pdf

Max 8 pages covering all
required sections

Demo videos

Isaac Sim viewer recording,
all levels labeled

Do not include virtual environments, Isaac Lab source, or large cache folders — keep the ZIP lean and reproducible.